

Is the Passive-Nominal Split Feasible?

A design gate for Predictive Interaction Realizability on the FR3

Working draft. Task 1 (the design gate), Task 2 (the merged controller), Task 3 (the authorization-vs-tracking sweep), and the root-cause analysis and fixes for what those found. Simulation only.

1. Decision summary

Merging `phri2`’s behaviour-realization separation with `impedance_residual`’s energy-authorized residual port requires `phri2` to acquire something it does not have: a **nominal**. The proposed route is the *passive-nominal split*,

$$F_{\text{cmd}} = F_{\text{nom}} + F_r, \quad F_{\text{nom}} = -K_0 e - D_0 v,$$

in which F_{nom} is a small behaviour-agnostic spring-damper that supplies a passivity floor and a harvestable dissipation term $v^\top D_0 v$, and the realization QP optimizes only the residual F_r that the tank then authorizes. The merged Lemma 1 that this route buys is worthless unless its first precondition is satisfiable: the **anchor**

$$a_\ell = \tau_{\text{base},\ell} + J_{v,\ell}^\top F_{\text{nom},\ell}, \quad |a_\ell| \leq \bar{\tau} \quad \text{at every tick,}$$

must fit inside the derated torque envelope — *and* the $\alpha \rightarrow 0$ fallback it degenerates to must still hold a sustained 20 N push inside the 0.06 m workspace bound. Those two requirements pull in opposite directions. This document reports whether they have a common solution.

They do, but only under one of the two envelopes the source papers use, and only in a sliver.

Torque envelope	Source	Feasible region	Verdict
$\rho \tau_{\text{max}}, \rho = 0.28$	<code>impedance_residual</code>	0 / 900 cells	NO-GO, structural
87/87/87/ 31.5 /12/12/12 N·m	<code>phri2</code>	33 / 900 cells , $k \in [360, 400]$ N/m, $d \in [21.3, 48.4]$ N·s/m	GO, thin

Under the $\rho = 0.28$ envelope no (K_0, D_0) works — not a tuning failure, a structural one (§5.1). Under `phri2`’s own envelope the split is viable at $K_0 = 380$ N/m, $D_0 = 29.1$ N·s/m, with every hard diagnostic sitting at 93-98 % of its limit (§5.2).

Three findings change what the merge can claim, and all three are load-bearing:

- The two papers’ “derated budget” stress cases are not interchangeable.** The synthesis note treats both as $\rho = 0.28$. They are not: `phri2`’s own realized torque peaks at **1.30×** the $\rho = 0.28$ cap, so `phri2`’s published FR3 benchmark does not even *run* inside `impedance_residual`’s envelope. The merge inherits whichever envelope it is written against, and the answer flips between them.
- The note’s four-diagnostic table admits false green cells.** Its row (1) is evaluated on a trajectory where F_{nom} is small, while its row (3) defines a displaced operating point the robot must actually hold. Checking each in its own place, 70 cells pass; requiring the precondition to hold on the *fallback itself* (row 1b, §4.3) leaves 33. Half the apparent region is an artifact of never asking one trajectory to satisfy both constraints.
- The $\alpha \rightarrow 0$ fallback transiently leaves the workspace box.** At the recommended cell the settled displacement is 55.8 mm against the 60 mm bound, but the transient peaks at **62.2 mm**. Gate on peak instead of settled displacement and the region drops from 33 cells to 6 — and the recommended cell is not among them.

The merged controller has since been implemented and run in closed loop (§7). That settles the gate’s main caveat and adds a fourth finding that outranks the other three:

- The split converts a hard actuator guarantee into a conditional one.** In `phri2` the QP’s torque constraint covers the *whole* command. Under the split the nominal sits outside the authorization loop — the 1 kHz servo can scale F_r but never F_{nom} — so once $|\tau_{\text{base}} + J_v^\top F_{\text{nom}}| > \bar{\tau}$ nothing in the fast loop can prevent the overrun. With only 2.4 % of joint 4’s

cap left as headroom, that happens at disturbance amplitudes well inside what `impedance_residual`’s own benchmark uses.

§8 root-causes all four and reports what can be done about them. In short: findings 1–3 are one problem, not three — diagnostic 3 forces K_0 above the desired impedance’s own stiffness, and every symptom is monotone in K_0 . Finding 4 is separate, and is fixed outright. §10 states the decision this hands to a human.

2. Why the split is needed at all

The two unsubmitted drafts are complementary halves of “predictive impedance under saturation”:

- `impedance_residual` holds a fixed physical impedance as nominal, lets a 100 Hz MPC propose an additive residual wrench, and authorizes that residual at 1 kHz against both a joint-torque envelope and an energy tank. Its central result is that the residual port must be re-authorized *between* manager updates.
- `phri2` emits a desired contact-port acceleration $a_{id} = C_\theta x + G_\theta F_h$ from a behaviour layer, realizes it in a receding-horizon QP under a horizon-wide torque constraint, and reports the deviation as $r = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}}$ rather than hiding it in clipping.

`phri2` §VIII names its own gap in as many words: *“the realization cost does not itself imply passivity ... A dissipativity constraint on the realized port variables, paired with an energy tank or passivity observer, is a natural extension.”* That extension is `impedance_residual`’s tank.

The graft does not work naively, for a structural reason. `impedance_residual`’s tank sits on a nominal τ_I that is always-feasible and natively passive, and it authorizes only the additive residual. `phri2` **has no nominal**. Its τ_{base} is feedforward plus orientation hold plus null-space centering, with no impedance term at all; the entire interaction behaviour lives inside F_{cmd} . Two consequences follow:

1. Scaling F_{cmd} down at a fast tick degenerates not to a passive spring-damper but to **zero interaction behaviour** — gravity compensation and posture only, a free-floating arm.
2. The tank’s measured advantage comes from harvesting the nominal damping $v^\top D_I v$. `phri2` has no harvestable damping term.

The passive-nominal split exists to fix both at once, at the cost of introducing (K_0, D_0) — and with them the question this document answers.

3. The merged fast re-authorization rule

For context, the rule the split is meant to enable. The QP supplies $F_{r,k}$ every 20 ms; the following runs every 1 kHz tick ℓ ($h = 1$ ms):

1. Recompute from the current $(q, \dot{q})$: $\tau_{\text{base},\ell}$, $J_{v,\ell}$, Λ_ℓ , and $F_{\text{nom},\ell} = -K_0 e_\ell - D_0 v_\ell$.
2. **Anchor** $a_\ell = \tau_{\text{base},\ell} + J_{v,\ell}^\top F_{\text{nom},\ell}$, with the precondition $|a_\ell| \leq \bar{\tau}$.
3. **Torque scale** $\alpha_{r,\ell} = \text{largest } \alpha \in [0, 1] \text{ with } |a_\ell + \alpha J_{v,\ell}^\top F_{r,k}| \leq \bar{\tau} \text{ per joint.}$
4. **Energy scale** $\alpha_{E,\ell} = 1$ if $\bar{F}_{r,\ell}^\top v_\ell \leq 0$, else $\min(1, (E_\ell - E_{\text{min}} + h v_\ell^\top D_0 v_\ell) / (h \bar{F}_{r,\ell}^\top v_\ell))$.
5. **Apply** and **ledger** as in `impedance_residual`.

Steps 1–2 are what §4–§6 gate. Steps 3–6 are implemented in `pir_controller.py` and measured in §7.

4. Method

4.1 One FR3 model, imported not rewritten

Every mass matrix, Jacobian, task-space inertia, bias torque, null-space projector, torque limit and scenario constant is imported from the existing studies (`simulation/fr3_mujoco.py` , `simulation/fr3_impedance.py` , `imp_reference/simulation/fr3_interaction_dynamics_mpc.py` , `imp_reference/simulation/run_fr3_experiments.py`). Nothing is re-derived: a second FR3 model would make this gate inconsistent with the papers it gates. The regression suite asserts the 0.06 m bound *is* `FR3MPCConfig.position_limit` rather than merely equalling it.

4.2 Scenario

phri2 's FR3 benchmark: the EE holds a fixed nominal pose while a smooth sustained 20 N push along $-z$ (raised-cosine ramp, `human_force_at`) would, under naive realization, drive it through the 0.06 m workspace bound. 6 s, 1 kHz servo, 50 Hz manager. Task-space inertia at the nominal pose is $\Lambda_{zz} = 4.56$ kg; the bias torque on joint 4 is 18.96 N·m before any control is applied at all.

Two torque envelopes are carried, because the two papers use different ones:

- `rho_0.28` — $0.28 \tau_{\max}$ on every joint (24.36 N·m on joints 1-4, 3.36 N·m on joints 5-7), `impedance_residual` 's continuous-envelope stress case.
- `derated_joint4` — stock FR3 limits with joint 4 derated 87 $\rightarrow$ 31.5 N·m, `phri2` 's own torque-activation stress case.

4.3 Diagnostics

Rows 1-4 follow the synthesis note's §13.2 table. Row 1b is added.

#	Quantity	Checks	Pass
1	$\max_t \ \tau_{\text{base}} + J_v^\top F_{\text{nom}}\ _\infty / \bar{\tau}$ on the <code>phri2</code> replay	anchor feasible	< 1
1b	the same ratio, peaked over the closed-loop $\alpha \rightarrow 0$ fallback run	anchor feasible <i>there too</i>	< 1
2	$\lambda_{\min}(K_0), \lambda_{\min}(D_0)$	passive	> 0
3	settled $\ e_z\ $ at $\alpha \rightarrow 0$ under the 20 N push	fallback holds	≤ 0.06 m
4	$\max_t \ J_v^\top F_{\text{nom}}\ _\infty / \bar{\tau}$	not budget-dominating	$< \beta = 0.5$

Why row 1b exists. Rows 1 and 3 are measured at two different operating points. Row 1 replays the trajectory `phri2` 's MPC actually produced, on which the EE stays near the nominal pose, so $F_{\text{nom}} = -K_0 e$ is small whatever K_0 is. Row 3 describes the *displaced* equilibrium the arm settles into once the residual is de-authorized, where by construction $K_0 e = F_h$. Merged Lemma 1 requires $|a_\ell| \leq \bar{\tau}$ at **every** tick, and the fallback's ticks are ticks. Row 1b closes that hole by peaking the same ratio over the fallback run itself — transient included, since Lemma 1's precondition has no exemption for overshoot. Omitting it more than doubles the apparent feasible region (§5.2).

Rows 1 and 4 are replay diagnostics: cheap, and exact *given* that the merged controller's total command tracks `phri2` 's. That assumption is the main caveat on this result and is stated as such in §7.

Row 3 is measured on a closed-loop MuJoCo run of the actual $\alpha \rightarrow 0$ law, $\tau = \tau_{\text{base}} + J_v^\top F_{\text{nom}}$, not from static algebra. The push hold is extended to 6 s and the displacement averaged over the final 1.5 s: at $k = 380$ N/m against $\Lambda_{zz} = 4.56$ kg the fallback spring has a 0.69 s natural period, so `phri2` 's own 2 s hold and a sub-period averaging window read an oscillation rather than an equilibrium — and biased the estimate in the flattering direction, making low-damping cells look *better* than they are.

4.4 Grid

$k \in [10, 1200]$ N/m log-spaced, $d \in [2, 60]$ N·s/m, densified through $k \in [300, 520]$ N/m and $d \in [8, 40]$ N·s/m: $36 \times 25 = 900$ cells. The note proposes $k \leq 400$ N/m and says to widen if the region sits at an edge. It does — row 3 alone forces $k \gtrsim 350$ N/m — and the densification is not cosmetic: on a 24×16 grid the region is ~ 40 N/m tall and the scan reports it **empty** (§6.3).

5. Results

5.1 `rho_0.28` : structurally infeasible

	value
τ_{base} alone, as a fraction of joint 4's cap	0.822
phri2 's own realized torque, as a fraction of this cap	1.300
cells passing row 1	197 / 900
cells passing row 1b	0 / 900
cells passing row 3	385 / 900
common region	0 cells
best worst-normalized use over the whole grid	1.238

Row 1b fails in **every cell**, with a grid-wide minimum of 1.144. This is not a tuning result. Holding a 20 N push at the fallback equilibrium requires $K_0 e = F_h$, so $J_v^\top F_{\text{nom}} = -J_v^\top F_h$ *regardless of* K_0 — the gains set where the arm sits, not how hard it must push back. On this pose that costs ~ 28.4 N·m on joint 4 asymptotically (§6.2), against a 24.36 N·m cap. No (K_0, D_0) can change that.

The second row of the table is the more consequential one. phri2 's published FR3 trajectory peaks at $1.30\times$ this envelope, so the envelope is not one the phri2 benchmark ever operated in. Carrying $\rho = 0.28$ into the merge, as the synthesis note assumes, does not merely tighten the design problem — it changes the scenario to one in which the source result does not exist.

5.2 derated_joint4 : feasible, thin

	value
τ_{base} alone, as a fraction of joint 4's cap	0.636
phri2 's own realized torque, as a fraction of this cap	1.005
cells passing row 1 / 1b / 3	577 / 702 / 385
common region	33 / 900 cells
extent	$k \in [360, 400]$ N/m, $d \in [21.3, 48.4]$ N·s/m
region if row 1b omitted	70 cells
region if row 3 gated on peak rather than settled	6 cells
region after $\beta = 0.5$	33 cells (row 4 never binds)

Recommended operating point. The note's literal criterion — minimize row 4 subject to rows 1–3 — selects $k = 360$ N/m, $d = 21.3$ N·s/m, which sits 0.4 % from violating row 1b. That is the wrong objective when the region is a sliver: row 4 is the only diagnostic the note does not hard-gate. The cell furthest from *every* hard constraint is the one to take:

$$K_0 = 380 \text{ N/m}, \quad D_0 = 29.1 \text{ N} \cdot \text{s/m}$$

diagnostic	value	limit	use
(1) replay anchor	0.977	1	98 %
(1b) fallback anchor	0.976	1	98 %
(3) settled $\ e_z\ $	55.8 mm	60 mm	93 %
(3') peak $\ e_z\ $	62.2 mm	60 mm	104 %
(4) budget fraction	0.348	$\beta = 0.5$	70 %

This is a damping ratio of $\zeta = 0.35$ at 1.45 Hz — a floor that is genuinely a floor, not a second behaviour layer.

The three-way squeeze. The overlay figure shows the region bounded below by row 3 (too soft and the fallback leaves the box), above by row 1 (too stiff and the anchor overruns joint 4 on the operating trajectory), and on the left by row 1b (too little damping and the fallback’s overshoot overruns joint 4). The note anticipated tensions (a) $\leftrightarrow$ (c) and (c) $\leftrightarrow$ (d). (d) never binds: row 4 ranges over 0.326–0.373 across the whole region, so β could be set anywhere above ~ 0.38 without changing a single cell. The binding third constraint is the one the note did not list — damping against the fallback transient.

5.3 Figures

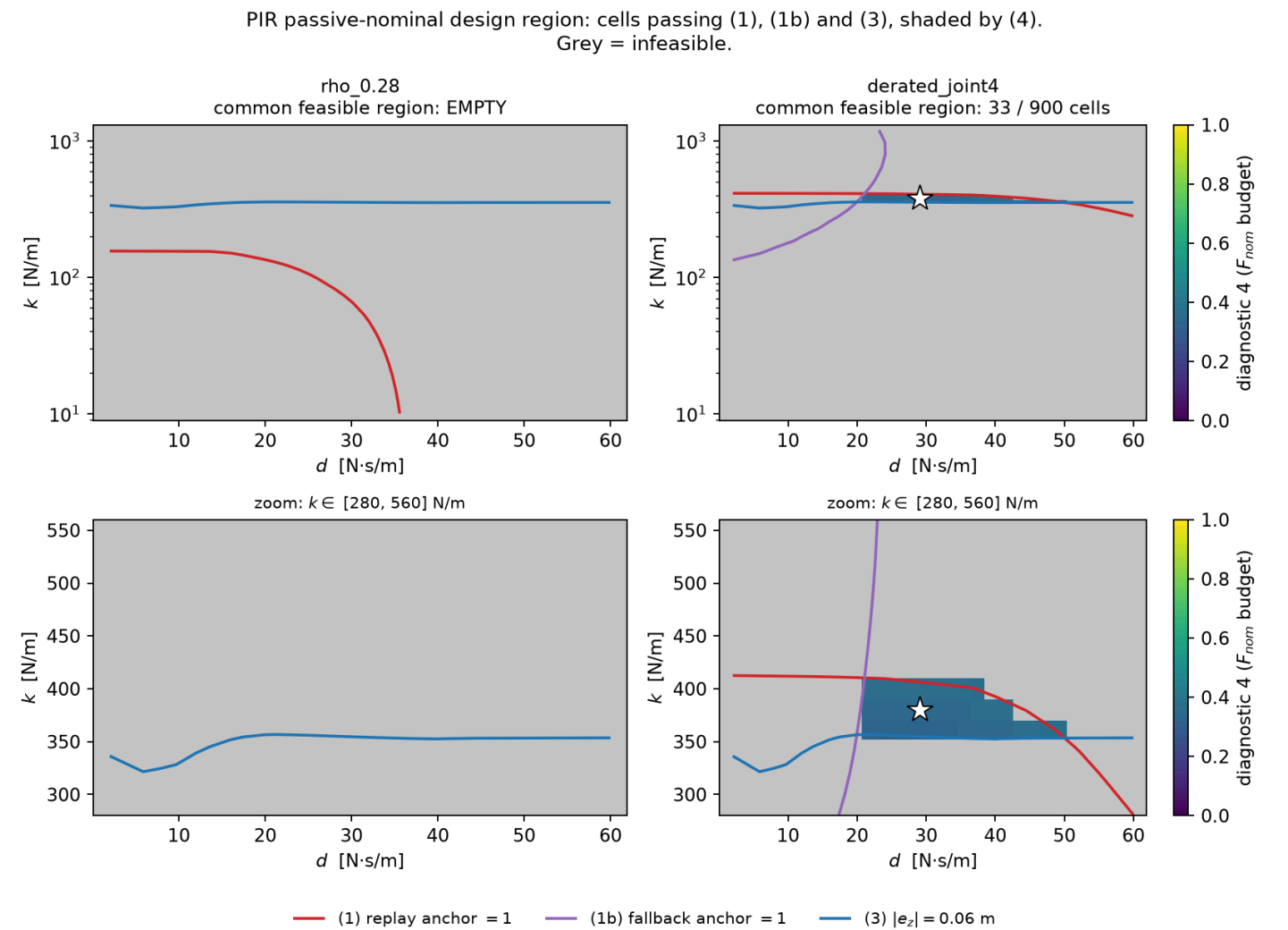


Figure 1 — the design region. Grey is infeasible. Left: the $\rho = 0.28$ envelope, empty everywhere. Right: `phr12`’s derated-joint-4 envelope, 33 of 900 cells, with the recommended operating point starred. The zoom row is not decoration — the region is ~ 40

N/m tall inside a 1190 N/m sweep, which is why a coarse grid reports it empty (§6.3). The three-way squeeze is visible in the lower-right panel: row (3) bounds it from below (too soft and the fallback leaves the box), row (1) from above (too stiff and the anchor overruns joint 4), row (1b) from the left (too little damping and the fallback’s overshoot overruns joint 4). The note anticipated the first two tensions; the third is the one it did not list.

file	content
pir_knot_scan_diag{1,1b,3,4}_{envelope}.png	the four diagnostic maps, per envelope
pir_knot_scan_overlay.png	composite, full range and zoomed, with the recommended cell marked
pir_knot_scan.json	grid, all diagnostic arrays, common_region_nonempty , both recommendations
pir_closed_loop_{scenario}_{envelope}.png/json	§7: the merged controller in closed loop
pir_e0_sweep.png/json	§7.4–7.5: authorization vs tracking, both tightenings active
pir_rootcause.png/json	§8.1: every symptom swept against K_0
pir_fixes.png/json	§8.3: candidate fixes scored on all four findings

6. Verification

A NO-GO on one envelope and a 33-cell GO on the other are both strong enough claims to be attacked separately (pir_verify.py , results/pir_verify.json).

6.1 Check A — equilibrium without integration

Rows 1b and 3 are read off a settled simulation, and “settled” is a judgement call for a lightly damped spring. Check A instead solves $\ddot{q}(q) = 0$ directly on MuJoCo’s own forward dynamics — no time integration at all — and compares. Across four cells the two methods agree to within **6.9 %** worst case (1.2 % at the well-damped end), all root-finds converged, all simulations flagged settled. The residual gap is the null-space posture spring, which makes the settled configuration mildly path-dependent; it is smaller than the 7 % margin the recommended cell carries on row 3, but not by much, which is part of why §7 calls that margin thin rather than adequate.

6.2 Check B — the fallback anchor does not depend on the gains

Sweeping k over [100,1200] N/m at the root-found equilibrium:

k [N/m]	$\ e_z\ $ [m]	$k \ e_z\ $ [N]	joint-4 anchor [N·m]
100	0.198	19.83	25.29
200	0.101	20.12	27.87
400	0.050	20.06	28.34
800	0.025	19.98	28.40
1200	0.017	19.95	28.39

$k \|e_z\|$ recovers the 20 N push to within 1 % across a 12× range of k , confirming $K_0 e = F_h$ and therefore that the equilibrium anchor is a property of the push and the pose, not of the gains. It converges to ~28.4 N·m — comfortably above the 24.36 N·m $\rho = 0.28$ cap and comfortably below the 31.5 N·m derated-joint-4 cap. That single number is why the two envelopes give opposite answers.

As a side result, the $k = 200$ row independently reproduces phri2 's own reported figure that a pure $K_d = 200$ impedance gives 0.10 m of static displacement against the 0.06 m bound.

6.3 Check C — grid refinement

Re-scanning $k \in [300, 500] \times d \in [8, 40]$ on a *linear* grid whose points deliberately do not coincide with the main scan's:

envelope	region	best worst-normalized use
rho_0.28	empty (0 / 99)	1.244
derated_joint4	14 / 99	0.977

Two things follow. The derated_joint4 verdict and its best cell are reproduced independently (0.977 against the main scan's 0.977). And the refinement is *necessary*: the original 24×16 grid reported derated_joint4 empty. The region is ~ 40 N/m tall inside a sweep spanning 1190 N/m, and a grid that samples it at two points misses it. Any re-run of this gate at a different pose, push direction or envelope must re-densify before trusting an empty answer.

6.4 Regression suite

test_pir_knot_scan.py , 12 tests, all passing. They pin the constants to their sources, check the anchor arithmetic against the per-tick matrix product, assert the zero-gain limit reduces the anchor to τ_{base} , verify the $K_0 e = F_h$ relation on the plant, check the averaging window exceeds the fallback spring's natural period, and assert the published grid stays dense enough through the feasible band.

7. The merged controller, implemented

Task 1's gate was a *feasibility* argument; §7 of its first draft owed a closed-loop measurement. pir_controller.py implements the merged rule of §3 and run_pir_closed_loop.py runs it on the FR3.

7.1 What the merge actually required

Three things changed relative to grafting the tank onto phri2 , and all three live in the QP:

- The nominal is closed-loop inside the horizon.** F_{nom} is state feedback, so the prediction runs on $A - BG_0$ with $G_0 = [K_0 \ D_0]$. Freezing F_{nom} across the horizon would let the QP plan against a nominal its own plan invalidates.
- The torque rows became state-dependent.** In phri2 , $\tau_k = \tau_{\text{base}} + J_v^T F_{\text{cmd},k}$ is affine in the decision variable alone. Here $\tau_k = \tau_{\text{base}} + J_v^T (-G_0 x_k + F_{r,k})$, and x_k is itself affine in every earlier residual, so each torque row carries the accumulated nominal reaction.
- The behaviour input and the disturbance model had to be split.** phri2 has one force signal driving both $a_{id} = Cx + GF_h$ and the plant prediction, which is sound only when every external force is interaction force. Running under a disturbance, they are different signals. control() now takes force_forecast and behaviour_forecast separately, defaulting to equal — which recovers phri2 exactly, and is asserted to. **This does not solve force misclassification**; deciding which measured force goes in which channel *is* that unsolved problem. It moves the assumption into a signature instead of hiding it in an equality.

One incidental gain: phri2 's QP has no principled infeasibility fallback and drops to its clipped reactive law. Here $F_r = 0$ is exactly the $\alpha \rightarrow 0$ passive nominal the Task 1 gate certified, so an infeasible solve degrades to a *certified* behaviour rather than an improvised one. (No solve was infeasible in any run reported here: 0 / 300 per run.)

7.2 Rows (1) and (4), re-measured in closed loop

This is the debt the first draft recorded. At $K_0 = 380$ N/m, $D_0 = 29.1$ N·s/m on phri2 's 20 N push:

	replay estimate (Task 1)	merged closed loop	gap
row (1) anchor ratio	0.9771	0.9762	0.09 %
row (4) budget ratio	0.3478	0.3474	0.11 %

The replay approximation holds. The gate’s 2 % margin on row (1) survives, and the GO verdict does not need revisiting on this scenario. A regression test pins the two together at 1 % so the gate cannot silently drift from the controller it gates.

7.3 Merged Lemma 1 and the four-term residual

Across every run in §7.4–7.5 (36 sweep points plus 8 closed-loop runs), with the precondition holding, the conclusion held: $|\tau_\ell| \leq \bar{\tau}$ at every one of 6000 ticks, with α_τ saturating the envelope exactly ($\max |\tau|/\bar{\tau} = 1.0000$) rather than overshooting it.

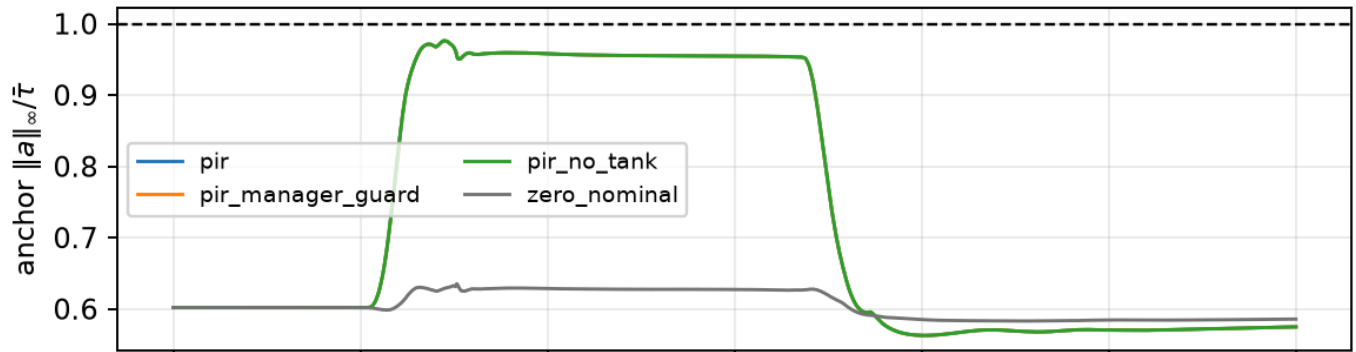
The four-term residual $r = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}} + r_{\text{auth}}$ closes **to machine precision** (4.4×10^{-16}) on every QP tick, with r_{con} split into its plan-level and execution-level halves:

$$r_{\text{con}} = \underbrace{a_{\text{con}} - a_{\text{unc}}}_{\text{QP, 50 Hz}} + \underbrace{\Lambda^{-1}(\alpha_\tau - 1)F_{r,k}}_{\text{servo, 1 kHz}}, \quad r_{\text{auth}} = \Lambda^{-1}(\alpha_E - 1)\bar{F}_{r,\ell}.$$

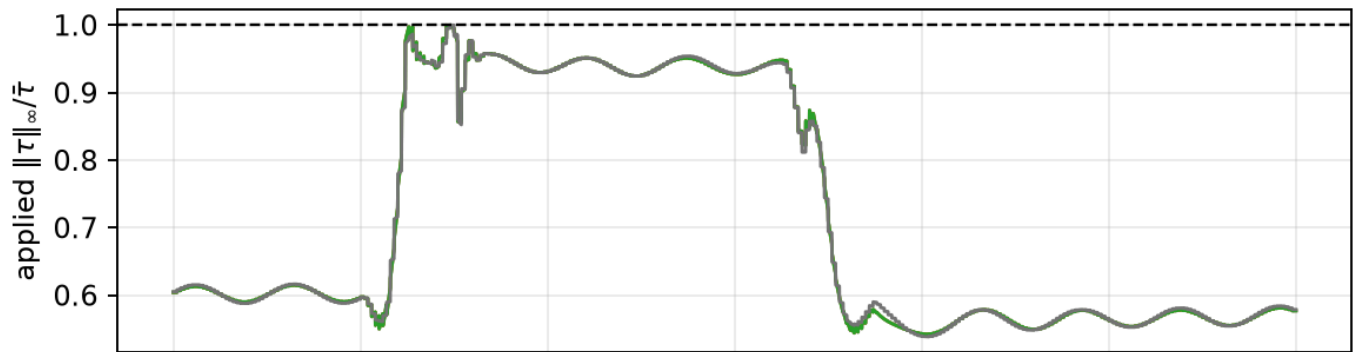
The closure is an identity, not a fit: the two servo terms telescope to $\Lambda^{-1}(F_{r,\ell} - F_{r,k})$, which is exactly the gap between what the QP proposed and what was applied. r_{auth} is identically zero whenever authorization never fires — it measures intervention, it is not an always-on correction.

PIR merged controller, scenario 'merged', envelope 'derated_joint4'

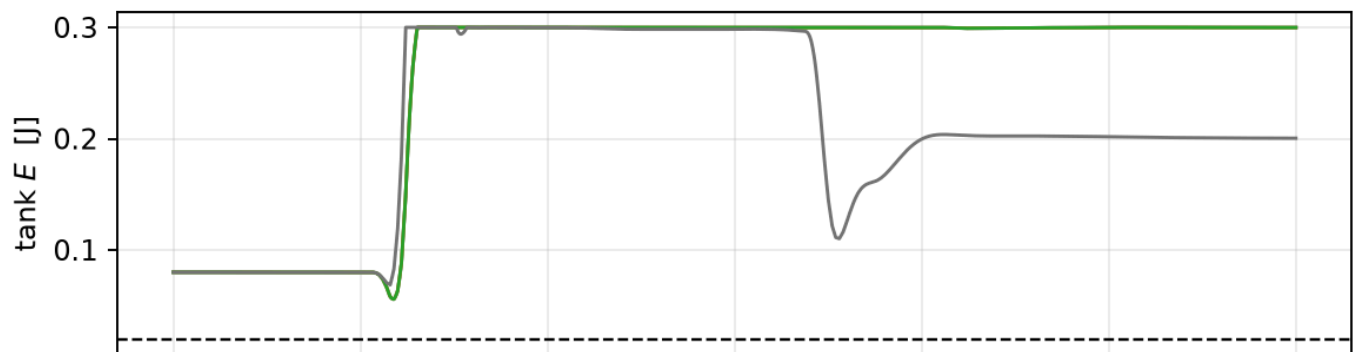
Merged Lemma 1's precondition (row 1), measured in closed loop



Merged Lemma 1's conclusion



energy tank vs its floor $E_{\min} = 0.02$ J



workspace excursion vs the 0.06 m bound

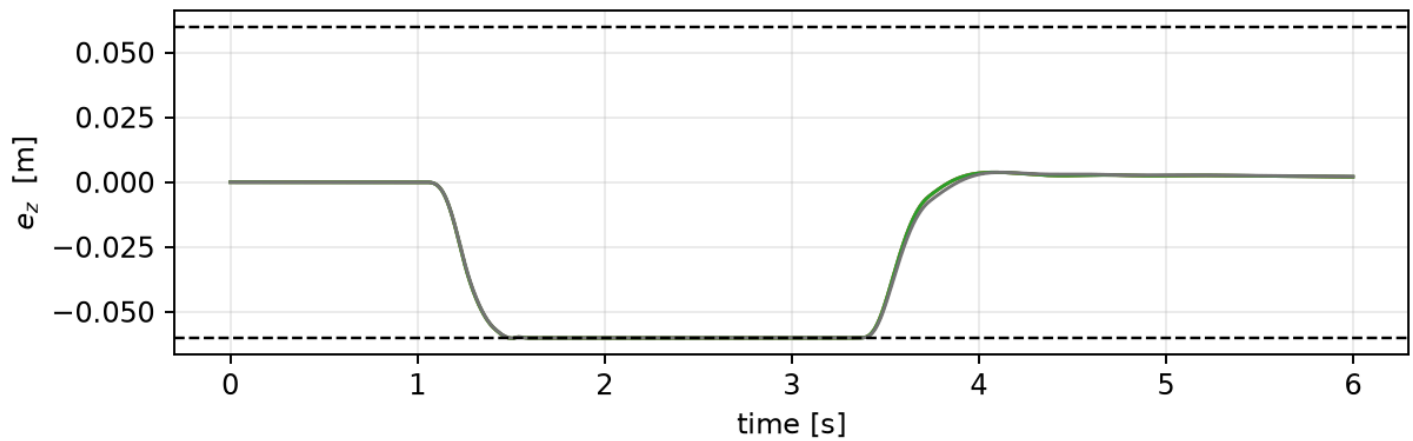


Figure 2 — the merged controller in closed loop, on the merged scenario at the recommended operating point. Top: Lemma 1’s precondition, the anchor ratio, sitting at 0.976 for the three PIR variants (they coincide here because authorization never fires at disturbance scale 1) against 0.635 for `zero_nominal`. That gap *is* \$7.6’s headroom collapse. Second panel: the conclusion — applied torque pinned at the envelope, never through it. Third: the tank, comfortably above its floor in this scenario, which is exactly why \$7.4 has to stress it deliberately. Bottom: the workspace excursion, slightly past the slack-relaxed 0.06 m box for every variant.

7.4 The passivity axis: `impedance_residual` ’s result transfers

Neither source benchmark exercises both axes. `phri2` ’s push is monotone and largely dissipative: the nominal harvests $v^\top D_0 v$ faster than the residual drains the tank, and $\alpha_E \equiv 1$ throughout. So the sweep runs on a **merged scenario** — that push plus `impedance_residual` ’s own rejectable disturbance (0.9/1.4/1.9 Hz sinusoids and a 12 N pulse placed deliberately between two 50 Hz manager ticks) — and sweeps E_0 toward its floor and the disturbance amplitude upward. 36 points, three variants:

variant	tank floor held	breach points	breached ticks	worst tank
<code>pir</code> (1 kHz re-authorization)	12 / 12	0	0	0.0200 J = $E_{\min}$
<code>pir_manager_guard</code> (20 ms, held)	4 / 12	8	2105	−0.188 J
<code>pir_no_tank</code>	4 / 12	8	5537	−1.473 J

`impedance_residual` ’s central result transfers to the merged port. Fast re-authorization holds the floor in every stressed configuration — and holds it *with equality*, $\min E = E_{\min}$ exactly, which is what the α_E construction guarantees. The manager-rate guard tracks it closely and breaches anyway, in 8 of the 12 configurations where the tank is loaded at all. The 20 ms staleness is the whole difference.

Authorization goes from silent to active as either axis is loaded: 0 % → 1.9 % of ticks as $E_0 \rightarrow E_{\min}$, and 0 % → 18.1 % of ticks as the disturbance scales 1 → 12.

PIR: authorization vs tracking with both tightenings active ($K_0 = 380.0$ N/m, $D_0 = 29.07$ N·s/m, envelope derated_joint4)

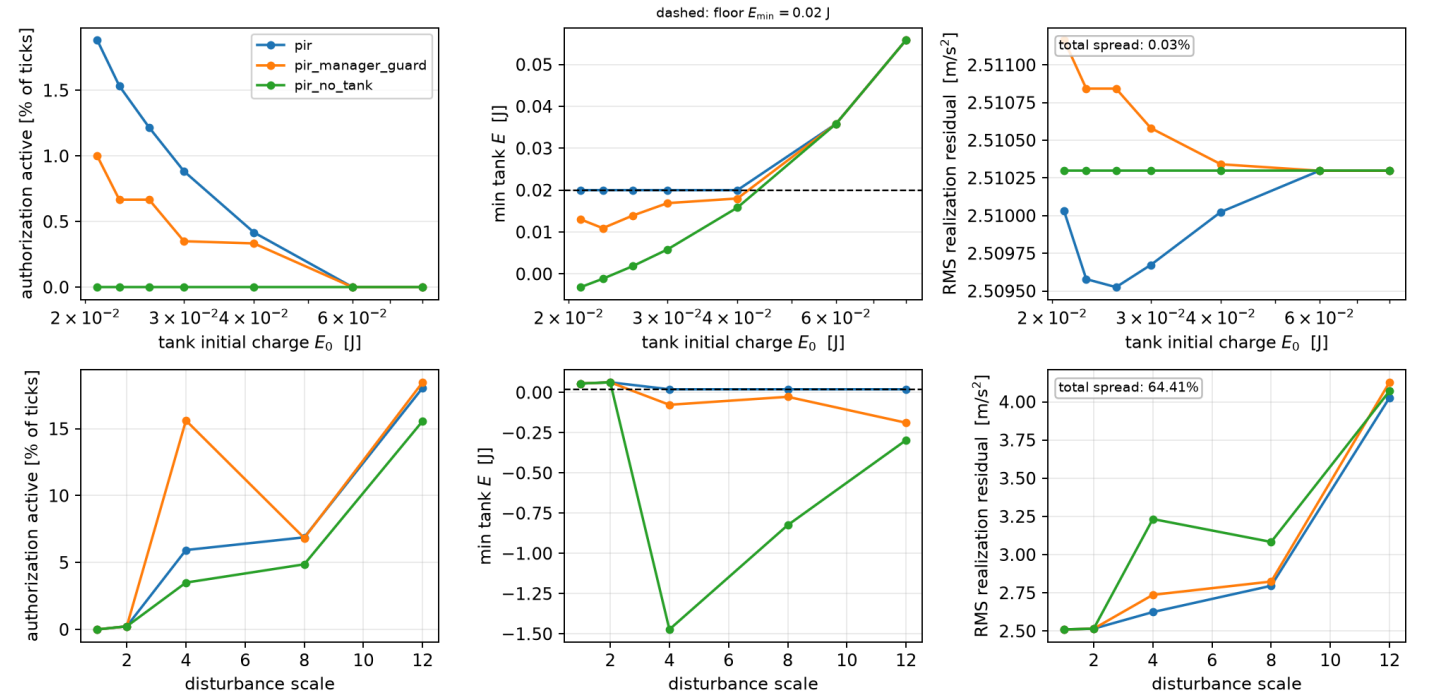


Figure 3 — authorization vs tracking, both tightenings active. Top row sweeps the tank’s initial charge toward its floor; bottom row sweeps the disturbance amplitude. The middle column is the result: `pir` (blue) sits *exactly* on $E_{\min}$, which is what the α_E construction guarantees; `pir_manager_guard` (orange) dips below it; `pir_no_tank` (green) reaches -1.47 J. The right column carries its own spread annotation because matplotlib autoscales it — on the E_0 axis the total variation is 0.03 %, i.e. flat (\$7.5), and only on the disturbance axis does a real trade appear.

7.5 The authorization-vs-tracking trade is *flatter*, not steeper

The synthesis note predicts the E_0 curve gets steeper after the merge, because the two tightenings compound in series. It gets flatter. Over the whole E_0 sweep, RMS realization residual varies by **0.03 %** ($2.5095 \rightarrow 2.5103$ m/s²) — against `impedance_residual`’s own $21.49 \rightarrow 15.98$ mm swing on its unmerged nominal.

The reason is the \$7.6 boundary, and it is not good news: the passive nominal has already absorbed the authority the tank would otherwise have taken away. De-authorizing a residual that is 16 % of the command costs almost nothing, because the other 84 % is a PD law the tank has no say over. A flat authorization-vs-tracking curve here means the passivity axis is *cheap* precisely because it is *weak*.

On the disturbance axis, where the residual does matter, the trade reappears: RMS spans $2.510 \rightarrow 4.026$ m/s² (60 %). Notably `pir` tracks *better* than `pir_no_tank` at disturbance scale 4 (2.624 vs 3.233 m/s²) — an unauthorized, energy-injecting residual makes realization worse, not better. That is one scenario at one seed and should not be leaned on.

7.6 The nominal dominates the command, and eats the headroom

At the only (K_0, D_0) the gate certifies:

	<code>pir</code>	<code>zero_nominal</code> ($K_0 = D_0 = 0$)
$\ F_{\text{nom}}\ $ RMS	14.05 N	0 N
$\ F_r\ $ RMS	2.11 N	13.48 N
residual’s share of the command	15.6 %	100 %
anchor headroom left for F_r	2.4 % of the cap	36.4 %

The note’s §7 predicted this qualitatively — “ K_0/D_0 large $\Rightarrow$ floor eats budget, a *new* double-tightening.” The measurement is worse than the phrasing suggests. $K_0 = 380$ N/m is not a small passivity floor sitting under the behaviour: it is **stiffer than the desired impedance** ($K_d = 200$ N/m), so the residual’s job is partly to make the arm *softer* than the floor. The merged controller at its certified operating point is a stiff PD with a 16 % predictive correction, and the residual’s torque headroom has collapsed **15-fold**, from 36.4 % of joint 4’s cap to 2.4 %.

7.7 Where the guarantee breaks

That 2.4 % headroom is what §1’s fourth finding cashes out. Sweeping the disturbance amplitude:

disturbance scale	max anchor ratio	ticks with infeasible anchor	max $\ \tau\ /\bar{\tau}$
4	0.9888	0 / 6000	1.0000
8	1.0074	39 / 6000	1.0074
12	1.0629	558 / 6000	1.0629

Two things to read here. First, the applied-torque overrun **equals the anchor overrun exactly** at both failing scales. That is not a coincidence: when the precondition fails, α_τ has already gone to zero and the fast layer has *no remaining authority* — the overrun is entirely the nominal’s, and scaling the residual cannot touch it. Second, `pir_no_tank` fails identically (1.0074 at scale 8), confirming this is a property of the nominal and has nothing to do with the tank.

So Merged Lemma 1 is sound and its precondition is doing real work: it is not a formality to be discharged once at design time. `phri2` guarantees the torque envelope unconditionally, because its QP owns the whole command. PIR guarantees it *conditional on the anchor being feasible*, and the Task 1 gate certified that condition against exactly one scenario. It does not survive a 4× larger disturbance. **This is the most important thing the implementation found, and it is a cost of the split that the synthesis note did not anticipate.**

8. Root cause, and what fixes it

Four findings is three too many if they share a cause. They do.

8.1 One constraint drives findings 1-3

Diagnostic 3 requires the $\alpha \rightarrow 0$ fallback to hold the whole 20 N push inside 0.06 m, i.e. $K_0 \geq |F_h|/0.06 = 333$ N/m. The behaviour the controller is supposed to render has $K_d = 200$ N/m. **The gate therefore forces the passivity floor to be stiffer than the behaviour it sits under**, and everything else follows. Sweeping K_0 at fixed damping ratio $\zeta = 0.35$ (`pir_rootcause.py`):

K_0 [N/m]	fallback $\ e_z\ $	(3) passes	anchor headroom	residual share	largest safe disturbance	E_0 trade
100	236 mm	no	26.4 %	72.6 %	12×	0.0068
150	154 mm	no	22.1 %	59.4 %	8×	0.0071
200 = K_d	112 mm	no	17.9 %	46.4 %	8×	0.0125
250	88 mm	no	13.7 %	34.1 %	8×	0.0358
300	72 mm	no	9.5 %	23.1 %	8×	0.0057
380 = certified	56 mm	yes	2.4 %	17.1 %	4×	0.0020
460	46 mm	yes	-1.2 %	39.1 %	0×	0.0016
600	34 mm	yes	-5.3 %	48.4 %	0×	0.0010

Every symptom is monotone in K_0 , every one of them gets worse as K_0 rises, and diagnostic 3 is the only diagnostic that wants K_0 large. At $K_0 = 460$ the headroom is already negative — the precondition fails at the *nominal* disturbance. The certified band is thin from both sides.

Is diagnostic 3's premise right? It asks the passivity *floor* alone to satisfy a workspace bound that the desired *behaviour* does not satisfy either: `phri2` 's own $K_d = 200$ gives 0.10 m of static displacement against the 0.06 m bound (reproduced here at 112 mm, §6.2), and `phri2` meets the bound through its **predictive layer**, not through its impedance. Requiring more of the floor than of the behaviour is a defensible safety choice, but it is a *choice*, not a physical constraint — and §8.3 prices it.

Root cause: every symptom is monotone in K_0 , and only diagnostic (3) wants K_0 large

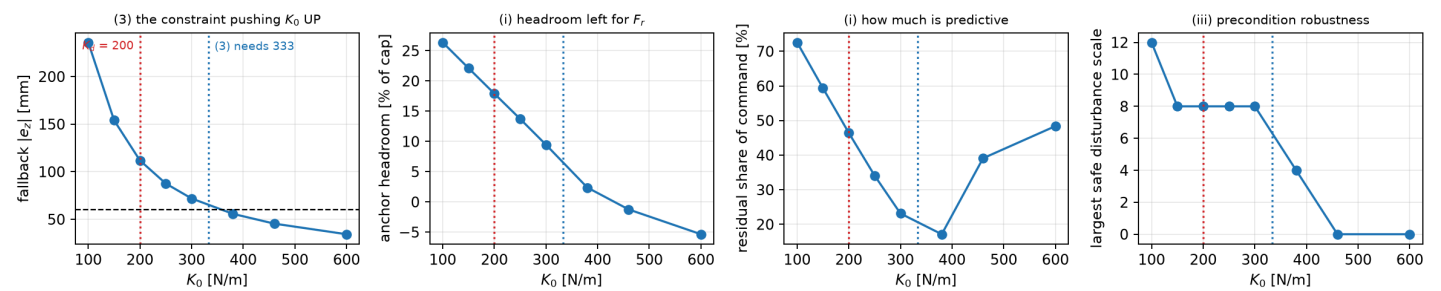


Figure 4 — the root cause. Red dotted line: the desired impedance’s own $K_d = 200$ N/m. Blue dotted line: the 333 N/m diagnostic 3 demands. The certified operating point sits to the right of both, which is why the headroom and the precondition robustness are near their worst there. Only the leftmost panel improves as K_0 rises.

8.2 Finding 4 is separate, and is fixed outright

Finding 4 — the fast layer has no authority over the nominal — is not a consequence of K_0 . It is structural: α_τ scales F_r only. The fix is to give the servo an α_{nom} as well, the largest scale keeping the anchor itself inside the box. Since $\alpha_{\text{nom}} = 0$ recovers τ_{base} , **the torque guarantee becomes unconditional whenever τ_{base} alone fits** — a far weaker and checkable condition (0.636 of the cap here) than “the anchor fits at every future tick”.

It is not free, and the first attempt traded one guarantee for the other. A time-varying spring gain is an energy term, and only one direction is dangerous: *softening* releases stored energy (the storage function’s $\frac{1}{2}\dot{a}e^\top K_0e$ term goes negative, which helps), while *re-stiffening* is an injection — at fixed e the robot suddenly pushes back harder without the human having done the work to store it. Charging re-stiffening to the tank and crediting softening nothing, unrestricted α_{nom} buys the torque envelope and **loses the tank floor** (−1.06 J at 16×): a transient repeatedly re-buys stiffness the port has not earned.

Forbidding the rise fixes it. With α_{nom} **monotone non-increasing**, both guarantees hold together out to 16× the source disturbance — against 4× for the certified baseline. The cost is a floor that does not recover its stiffness within an episode; a deployed system would reset it per contact, which this benchmark does not exercise.

8.3 Candidates, scored on all four findings

pir_fixes.py , all on the merged scenario, all against the same probes:

candidate	fallback	(3)	headroom	residual share	torque OK to	+ tank floor to	E_0 trade	max $\ e_z\ $ at 12×
certified $K_0 = 380$	56 mm	✓	2.4 %	17.1 %	4×	4×	0.0020	79 mm
soft_nominal $K_0 = K_d$	110 mm	✗	16.3 %	46.0 %	8×	8×	0.0070	—
anisotropic	56 mm	✓	2.2 %	20.0 %	2×	2×	0.0003	—
nominal_auth	56 mm	✓	2.4 %	17.1 %	16×	6×	0.0020	—
nominal_auth_mono	56 mm	✓	2.4 %	17.1 %	16×	16×	0.0020	—
soft_plus_auth	110 mm	✗	16.3 %	46.0 %	16×	8×	0.0070	—
soft_plus_mono	110 mm	✗	16.3 %	46.0 %	16×	16×	0.0070	160 mm

Candidate fixes for the three Section 7 findings

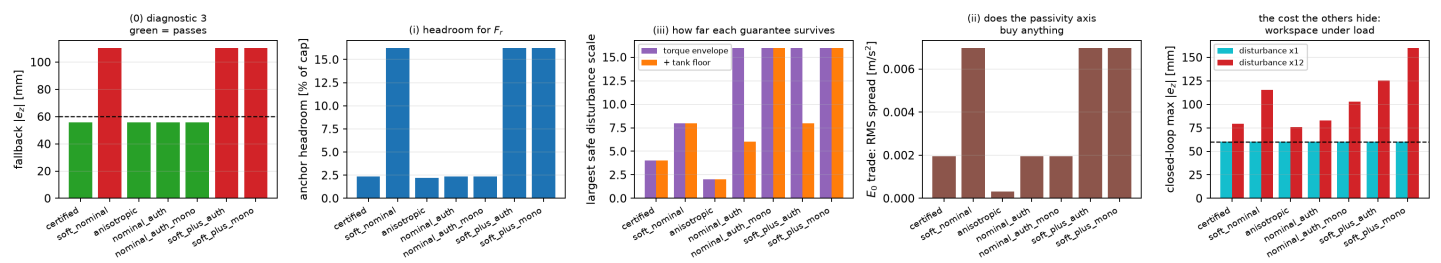


Figure 5 — candidate fixes. The rightmost panel is the one that changes the recommendation: at nominal load every candidate holds the workspace box identically, and only under a 12× disturbance does the soft nominal’s real cost appear. Without that panel `soft_nominal` looks free.

Three things to take from this.

`nominal_auth_mono` **is a pure win and should be adopted.** It quadruples the range over which *both* guarantees survive, costs nothing on any other metric, changes nothing at all while the anchor fits (asserted by test), and keeps diagnostic 3. There is no argument against it in these measurements.

The anisotropic refinement — the plan’s own deferred “later refinement” — does not work here, and slightly hurts. Headroom falls to 2.2 % and the safe disturbance range halves to 2×. The reason is visible in the trajectory: the displacement is 8:1 dominated by the push axis (max $|e| = [7.5, 0.3, 60.3]$ mm), so softening the off-axis gains barely reduces $\|J_v^\top F_{\text{nom}}\|_\infty$ while removing the off-axis restoring force that was helping keep the arm near the pose. A negative result, but a clean one: the obvious lever is the wrong lever.

Lowering K_0 is not free, and the fallback bound is not the only thing it buys. At nominal load every candidate holds the box identically (60.2 mm), which makes `soft_nominal` look costless. Under load it is not: at 12× disturbance the certified nominal holds 79 mm while `soft_plus_mono` reaches **160 mm**, and its RMS realization residual is worse too (5.60 vs 4.03 m/s²). So the real trade for findings 1–3 is not “56 mm vs 110 mm in a fallback that may never happen” — it is **workspace containment under disturbance**, which is a live property. That is a genuine engineering decision, and it is the human’s to make, not the scan’s.

8.4 What this does and does not settle

Findings 1–3 are one problem with a known knob and a priced trade. Finding 4 is solved. What remains open is the same thing §7.7 pointed at: even with α_{nom} , the guarantee is “torque legal provided τ_{base} fits”, and nothing here defends *that*. On this FR3 pose τ_{base} uses 0.636 of joint 4’s cap with no control authority applied at all, so the margin is real but finite, and it is a pose property — which is why re-running the gate at a pose that does not load joint 4 remains the highest-value untried experiment (§10).

9. What this does not show

- **The merged controller is not implemented.** Discharged in §7.2: the replay estimate and the merged closed loop agree to 0.1 % on both rows, and a regression test now pins them together.
- **The anchor’s feasibility does not survive a larger scenario.** Fixed in §8.2 by `nominal_auth_mono`, which holds both guarantees to 16×. What is *not* fixed is the condition underneath it: everything still assumes τ_{base} alone fits the envelope, which is a property of the pose and is not defended anywhere.
- **The fix’s cost is not fully characterised.** `nominal_auth_mono` ratchets the floor’s stiffness down and never recovers it within a run. A 6 s benchmark does not show what that does over minutes of interaction, and the per-contact reset a deployed system would need is not implemented or tested.
- **The K_0 trade is priced on one disturbance profile.** §8.3’s workspace numbers at 12× come from `impedance_residual`’s own rejectable force scaled up, at one seed. That is a stress test, not a distribution.
- **The fallback leaves the workspace box in transient.** 62.2 mm peak against a 60 mm bound. `phri2`’s box is slack-relaxed rather than hard, so this is not a constraint violation in its formulation — but it means the $\alpha \rightarrow 0$ guarantee is “settles inside the box”, not “stays inside it”. If the bound is to be read as hard, the region is 6 cells, not 33, and the recommended cell is outside it.
- **One pose, one push direction, isotropic gains.** The whole gate is evaluated at `phri2`’s nominal hold pose under a $-z$ push, with $K_0 = kI$, $D_0 = dI$. Joint 4 is the binding joint in every single cell, and it is binding largely because of an 18.96 N·m bias torque at that specific configuration. An anisotropic (K_0, D_0) — stiffer along the push axis, softer elsewhere — is the obvious first lever if the region needs widening, and the note explicitly defers it. A different pose could move the verdict in either direction.
- **Recursive feasibility is untouched**, in both source papers and here. The combined torque-envelope + energy-floor + workspace-slack feasibility question is harder jointly than separately and cannot be self-certified.
- **The force-misclassification pillar is untouched.** The tank meters $F_r^\top v$; if F_h leaks into $\hat{d}$, or F_e is folded into the disturbance, the power sign is wrong and every certificate here stays green while the arm pushes a mislabelled wall. **No green cell in this document implies anything about that pillar.**

- **Double tightening is not measured.** `phri2` 's horizon-wide torque tightening and `impedance_residual` 's tank tightening compound in series. That curve is Task 3 and is expected to get steeper after the merge.
- Simulation only, translational only, affine memoryless behaviour class.

10. Decision gate

Per §13.3 of the synthesis note, this is where autonomous work stops.

`common_region_nonempty == true` under `phri2` 's derated-joint-4 envelope, the merged controller is implemented, Merged Lemma 1 holds in closed loop and the four-term residual closes exactly. What is left is not a coding decision.

§8 has changed what is being decided. One of the four findings is now fixed outright and should simply be adopted; the other three collapse into a single priced trade. What is left for a human:

Decision 0 — adopt `nominal_auth_mono` ? I recommend yes. It quadruples the range over which both guarantees survive, is provably inert while the anchor fits, and costs nothing measurable. The only reservation is §9's: its stiffness ratchet is untested beyond a 6 s run, so it needs a per-contact reset before hardware.

Decision 1 — how much workspace containment under disturbance is the predictive layer's authority worth? That is the real form of findings 1-3. $K_0 = K_d = 200$ buys $6.8\times$ the anchor headroom, $2.7\times$ the residual share and a $3.5\times$ steeper E_0 trade, and pays with 160 mm of excursion at $12\times$ disturbance against 79 mm. Neither end is obviously right.

Decision 2 — which envelope does the merged paper claim? Unchanged from before the implementation, and still the question that decides whether any of this is publishable as one paper:

Which envelope does the merged paper claim? The two are not interchangeable and they give opposite answers. Three options, in the order I would rank them:

1. **Write the merge against `phri2` 's derated-joint-4 envelope.** $(K_0, D_0) = (380, 29.1)$, region of 33 cells, margins of 2-7 %. Honest, and the source benchmark actually runs there. Costs: the tank's numbers from `impedance_residual` were measured under $\rho = 0.28$ and would have to be re-derived, and the margins are thin enough that §7's first bullet could consume them.
2. **Keep $\rho = 0.28$ and abandon the passive-nominal split,** retreating to "dissipativity directly on the realized port, no nominal" — a different and harder design, and a separate planning decision. §5.1 says this is forced, not a preference: under $\rho = 0.28$ no nominal exists.
3. **Change the scenario** — a different nominal pose, a lower push magnitude, or a push direction that does not load joint 4 — and re-run this gate. Cheap to try (the scan is ~ 19 min on 4 cores) and it attacks the actual binding quantity, which is an 18.96 N·m bias torque at one configuration rather than anything about impedance.

Option 3 has gained the most weight. The binding quantity is still an 18.96 N·m bias torque at one configuration, §8.4 shows that even the fixed controller rests on π_{base} fitting, and §8.3 shows the only lever that widens the headroom without the fix costs workspace containment. A pose or push direction that does not load joint 4 would relieve all of it at once, and nothing measured so far tells us whether it would. **It is the highest-value untried experiment and it is cheap** — the gate is ~ 19 min and the fix comparison ~ 5 min. I have not chosen among these; the measurements say what each costs.

Owed before any claim

- ☐ **Human:** decisions 0, 1 and 2 above.
- ☒ ~~Root cause the three Section 7 findings~~ — §8.1. One constraint, monotone in K_0 .
- ☒ ~~Fix the precondition failure~~ — §8.2, `nominal_auth_mono`. Both guarantees to $16\times$.
- ☒ ~~Try anisotropic (K_0, D_0)~~ — §8.3. It does not help here and slightly hurts; the displacement is 8:1 push-axis dominated.
- ☐ **Re-run the gate at a pose that does not load joint 4** (see above).
- ☒ ~~Implement the merged controller and re-measure rows 1 and 4 in its own closed loop~~ — §7.1-7.2. Replay and closed loop agree to 0.1 %.
- ☒ ~~Task 3: re-sweep the authorization vs tracking curve (E_0) with both tightenings active~~ — §7.4-7.5. The trade is **flatter**, not steeper, and §7.5 says why that is a bad sign rather than a good one.

- **Task 2, the proof:** write merged Lemma 1 (per-joint α_r closed form, $\alpha_r \cdot \alpha_E$ on-segment lemma) and merged Proposition 1 at submittable granularity. The implementation supplies the operating point and confirms both halves empirically; the write-up must state the precondition as a **standing hypothesis that can fail at run time** (§7.7), not as a design-time check.
 - ☑ ~~Decide what the controller does when the precondition fails~~ — §8.2. Folding the nominal into the authorization loop works, and the passivity cost it first appeared to carry is removed by making α_{nom} monotone. The escalation route back into certified-realizability is still the answer for the residual condition §8.4 names (τ_{base} itself), which no local fix defends.
 - Re-run the Task 1 gate at a pose/push direction that does not load joint 4, and check whether the headroom problem is FR3-pose-specific.
 - Implement and test the per-contact reset `nominal_auth_mono` needs, and characterise the ratchet over runs longer than 6 s.
 - Resolve whether the 0.06 m bound is hard (region = 6 cells) or slack-relaxed (region = 33 cells) — and note that §8.1 makes this the same question as decision 1.
 - Explicitly disclaim the force-misclassification pillar in whatever is written. §7.1’s channel split makes the assumption visible; it does not discharge it.
-

11. Reproducing

```

cd simulation
python3 pir_knot_scan.py           # ~19 min on 4 cores (Task 1 gate)
python3 pir_knot_scan.py --rescore # re-derive verdicts/figures from stored maps
python3 pir_verify.py             # ~3 min (gate verification)
python3 run_pir_closed_loop.py --scenario push # ~35 s (Section 7.2, 7.6)
python3 run_pir_closed_loop.py --scenario merged # ~35 s (Section 7.3, 7.4)
python3 pir_e0_sweep.py           # ~6 min (Section 7.4, 7.5)
python3 pir_rootcause.py          # ~4 min (Section 8.1)
python3 pir_fixes.py              # ~5 min (Section 8.3)
python3 -m pytest test_pir_knot_scan.py test_pir_controller.py -q

```

MuJoCo mesh assets are gitignored repo-wide and must be fetched from MuJoCo Menagerie into

```

../simulation/models/franka_fr3/assets/ . Pinned dependency versions are in ../impedance/simulation/requirements.txt ;
mujoco==3.10.0 matters, since 3.13 renamed MjData.qM and fr3_mujoco.py reads it.

```